

Defining the concept of manner: An attempt to order chaos

Dejan Stosic

University of Toulouse Jean Jaurès, Laboratory CLLE

CNRS & UT2J, France

The concept of manner is extensively used in many fields of linguistics. It also appears to ordinary speakers, students and grammarians as an intuitively clear and meaningful notion that does not need any precise definition. Some scholars even claim that manner belongs to a limited set of primitive semantic categories that cannot be further broken down into more specific sub-units, and which it is neither necessary, nor possible, to define. Nevertheless, such an assessment is a significant obstacle when attempting to provide fine-grained analyses of linguistic phenomena pertaining to manner. The aim of this paper is to propose a more comprehensive definition of the concept of manner by bringing to the fore some basic mechanisms underlying the occurrence of the manner component in utterances, be it at the syntactic, lexical, grammatical or morphological level.

Keywords: manner, semantics, manner adjuncts, evaluative morphology

Manner is a pervasive linguistic phenomenon, present in almost all types of language productions, from everyday communication, literature, and media to a variety of specialized, institutional, private, cultural, scientific, economic and other discourses. Manner can thus be considered as structuring almost all conceptual domains, and as one of the main semantic categories. This pervasiveness is closely linked to the fact that languages generally display a large variety of means and strategies for expressing this crucial semantic component. From a descriptive point of view, the concept of manner is widely used in many fields of linguistics, especially in syntax, morphosyntax and lexical semantics. Despite extensive research on certain kinds of means of expressing manner in several languages, it must be acknowledged that little is known about how the manner component is processed in utterances, and what semantic values have to be included in it. Moreover, one cannot fail to notice that, in linguistics, this notion is always used on an intuitive basis and no serious attempt has been made to outline a genuine definition of manner. It is usually considered as a specific way of performing an action (e.g. *to live* vs *to live comfortably*, *to live in a carefree manner*). The two major problems of such a definition are, first, that it is circular because it uses the very similar and ill-defined term “way”, and second, that it can embrace a large variety of meanings (e.g. *to live dangerously*, *to live luxuriously*, *to live anonymously*). The latter issue is precisely what makes manner so heterogeneous in the eyes of many scholars (see Guimier 1996), and dooms to failure any attempt at formulating a coherent and suitable definition for this concept.

The aim of this paper is to provide a more comprehensive semantic definition of manner, which is needed when studying the linguistic phenomena linked to this concept more deeply and more accurately (cf. Stosic 2011). The paper is structured in four sections. In the first section, I review various ways of dealing with manner in contemporary linguistics. The second section outlines an onomasiological approach to manner by providing a

survey of five main types of linguistic means involved in its expression. This allows me to introduce, in the third section, the principal mechanisms of how manner interpretation is processed in utterances, as well as a more precise, holistic definition of this semantic concept. The fourth section further elaborates the definition by articulating manner with some related concepts such as instrument, means, intensity, and so on. French will be my main language of demonstration in this paper, but data from several other languages will also be used for the purposes of illustration.

1. Manner: an ill-defined concept

The way manner has been studied in French linguistics is a good example of the approach that was generally used during the 20th century to describe this notion. In what follows, I briefly discuss the main tendencies of the previous research dealing with manner, which remains an ill-defined concept despite the numerous studies carried out on it.

1.1. Sharpening the focus on manner adjuncts

A survey of the previous research on the expression of manner clearly indicates that manner adjuncts / adverbials have been the main focus of most of the studies dealing with this concept. This general tendency is well illustrated by numerous investigations by French linguists in the second half of the twentieth century who argued that using manner adjuncts / adverbials is the most common way of expressing manner in French (see, among others, Nilsson-Ehle 1941; Gross 1990; Nøjgaard 1992–1995; Guimier 1996; Molinier and Lévrier 2000). During several decades, significant efforts were made to formulate appropriate criteria for delineating the boundaries of such a complex and, both syntactically and semantically, fuzzy category. Since, to a certain extent, a great number of adverbs exhibit a strong affinity with

manner from morphological, semantic and syntactic points of view, many scholars have additionally reduced the encoding of manner to the sole category of adverbs (see Gary-Prieur 1982 for a more precise development). For instance, in example (1), one can easily observe first the presence of a special morphological marking of manner in the very form of the adverb (*-ment* in French, *-ly* in English), second the fact that the adverb conveys a particular way of performing the process of transforming the codes of dance, and third that, syntactically, the adverb is integrated in the VP and functions as a modifier of the verb. These converging properties have led certain linguists to claim a one-to-one relationship between adverbs such as *profondément* in French (or *profoundly* in English) and the concept of manner.

- (1) *Artiste majeur de la scène internationale, Jérôme Bel a **profondément** bouleversé les codes de la danse, classique et contemporaine.* (Opéra de Paris Site)
 ‘Jérôme Bel, one of the greatest artists on the international scene, has **profoundly** transformed the codes of classical and contemporary dance.’

Adverbs thus appeared to be the natural candidates to represent manner adjuncts, and they even attained the status of prototypical expressions of manner (see Gary-Prieur 1982; Virtanen 2008). But while most adverbs contribute to expressing manner (as *profondément* and *profoundly* do), they also entail a wide range of other values, and there is no comprehensive study comparing the use of adverbs with the use of other types of manner adjuncts in order to establish their actual status in this paradigm (see, however, Hasselgård 2010; Stosic 2013a for such attempts). As a result, the focus shifted from looking for criteria to identify manner adjuncts to trying to define formal criteria for opposing adverbs of manner to those conveying other values such as quantity, intensity, modality, and so on.

Despite many fruitful descriptive results, it must be acknowledged that this approach has several limitations (see Stosic 2011 and Moline & Stosic 2016 for further developments). Two of them are particularly serious when one wants to draw the outlines of the subclass of manner adjuncts / adverbials. The first limitation of this type of approach is the lack of generalizability of the findings in light of highly diverging sub-classifications of adverbs from one study to another. As a result, significant differences are found in what the proposed classifications consider as adverbs of manner, as can be clearly seen when comparing, for instance, Nilsson-Ehle's (1941), Molinier's (1990), Nølke's (1990), Nøjgaard's (1995), Guimier's (1996), Molinier and Lévrier's (2000), and Bonami *et al.*'s (2004) proposals. It is worth noting that all these authors used formal criteria for classifying adverbs, but few of them applied the same criteria. The second limitation arises when one tries to extend these criteria beyond the boundaries of adverbs to other syntactic structures involved in the expression of manner, such as prepositional phrases, gerunds, some types of subordinate clauses, and so on (see 2.1. below). This is what Moline (2011) showed by comparing Nøjgaard's (1995) and Molinier and Lévrier's (2000) sets of criteria and by trying to apply them to other types of manner adjuncts. The author concluded as follows:

“The criteria examined (...) seem to be relevant only within the framework of the classifications in which they were conceived, and they do not enable the paradigm of “manner adjuncts” to be built: none of them is specific for this type of adjuncts, and none of them applies to all these adjuncts. Using many of these criteria simultaneously seems hardly more useful.” (Moline, 2011: 94)¹

¹ « Les critères examinés (...) ne semblent pertinents que dans le cadre des typologies dans lesquelles ils ont été conçus, et ne permettent pas de construire un paradigme des « compléments de manière » : aucun d’eux n’est spécifique de ce type

However, Moline stresses that this in no way means that manner adjuncts do not exist; she rather calls into question the reliability of the sole use of syntactic manipulations to identify manner adjuncts. It thus appears that a broad range of both semantic and syntactic constraints specific to a particular formal type of expressions or even to a particular item make formal criteria ineffective as a means of identifying the sub-class of manner adjuncts.

This brief survey suggests that the excessive focus on essentially syntactic issues when dealing with manner has proved to be a somewhat misleading way of characterizing this semantic notion.

1.2. To define or not to define?

Basically, there are two ways of approaching the problem of defining the concept of manner. The first is to avoid it by using various intuitive assessments close to what can be found in dictionaries. In this case, manner is considered as corresponding to:

- “a particular aspect of a process, action or state” (*TLFi*),
- “the idea of quality applied to essentially verbal notions” (Sechehaye 1926),
- “the quality applied to process” (Rémi-Giraud 1998),
- “a heterogeneous value involved in various domains” (Guimier 1996),
- “the way in which the action the verbs denote is performed” (Levin and Rappaport Hovav 1992)

Such assessments about manner are consistently unable to accommodate certain linguistic data due to their lack of discriminating potential or to their

de complément et aucun d’eux ne s’applique à l’ensemble de ces compléments. L’utilisation simultanée de plusieurs critères (...) ne paraît guère plus utile.» (Moline, 2011: 94)

overly restrictive scope, so that the intuitive use of the concept of manner in language description is highly problematic. For instance, one can consider that quantifying processes, actions or states (e.g. *to help a lot*, *to talk a long time*, *to be too dangerous*) is also a way of describing a particular aspect of these entities, whereas manner qualification must be clearly distinguished from quantification. Likewise, saying that manner is the quality applied to processes, or verbal notions, requires a clear understanding of the notion of quality, which is generally lacking. Due to its fuzziness, the concept of manner thus appears both too powerful and not refined enough to accurately capture subtle differences between different kinds of linguistic means involving this value, which, although assumed to be uniform, turns out to be extremely heterogeneous. Whatever the language of the study, the heterogeneity of manner is regularly observed at the level of the linguistic forms used to express it (cf. Beavers *et al.* 2010, Hasselgård 2010, Moline & Stosic 2016), but also at the semantic level (cf. Melis 1983, Guimier 1996, Fellbaum 2002, Stosic 2009, Iacobini & Vergaro 2012, Slobin *et al.* 2014).

The second way of solving the problem of definition consists in refuting either the possibility or the necessity of providing any precise definition of the concept of manner. This view is supported by the claim that manner is one of the main semantic and conceptual categories that structures almost all conceptual domains. As such, manner is included in a small set of basic – and hence irreducible – ontological categories, and placed on the same level as HUMAN, THING, PLACE, ACTION, QUANTITY, REASON, and so on (see Jackendoff 1983; Heine *et al.* 1991; Haspelmath 1997; Le Goffic 2002). Consequently, defining manner should be *a priori* impossible. While the study of a range of linguistic data involving the manner component suggests that manner deserves to be considered as one of the fundamental ontological

categories², there can be no assurance that it could not be reduced to more atomic components and mechanisms. In any case, arguing that both *to walk* and *to run* involve the manner component, while true, obviously does not provide a satisfactory explanation: the two verbs arguably express different ways of moving. So, the manner component they convey must be based on disparate features.

1.3. A quick peek into the conceptual neighbourhood

Another major difficulty in using manner as an analytical and descriptive semantic category in linguistics lies in the great complexity of relationships that manner enters into with a set of neighbouring concepts such as quality, instrument (2), means (3), intensity, comparison (4), and so on. As shown in the following examples, it is debatable whether the highlighted constituents involve manner or another contiguous notion. In the vast majority of such cases, both manner and another semantic value seem to co-occur in the interpretation of utterances, in an inseparable way.

- (2) He then signed **with his hand** for Heyward to retire [...] (J. F. Cooper, *The Last of the Mohicans*, 1826)
- (3) The other acknowledged the pacific compliment **by a gesture of the hand**, and remained silent. (J. F. Cooper, *The Last of the Mohicans*, 1826)
- (4) She told me yesterday that her husband does not treat her **as she wishes**.

² Ontological categories are meant here as top-level systematizations of our knowledge of the world constructed, in large measure, based on linguistic input, rather than as essential characteristics of the world itself (see among others Westerhoff 2005).

For example, in (2), the PP *with his hand* can be interpreted as conveying both instrument and manner. In example (3), one can question by saying both (i) *how did the other acknowledge the compliment?* and (ii) *by what means did the other acknowledge the compliment.* In (4), it is very difficult to separate comparison and manner because the subordinate comparative clause *as she wishes* also expresses a particular way of treating the person. A robust definition of manner must allow either its delineation from these contiguous notions or its possible articulation with them. I will come back this issue in the fourth section below.

This short survey of the main tendencies in studying means of expressing manner emphasises some major weaknesses of the way this concept is generally approached in linguistics. As a result, and due to the excessive focus on syntactic aspects, the lack of definition, and the fuzzy boundaries between manner and some other semantic notions, the concept of manner lacks sufficient explanatory power (see Stosic 2011 and Moline & Stosic 2016 for a fuller discussion). In what follows, I shall sketch out an alternative way of approaching manner, based on my previous research.

2. Towards an onomasiological approach to manner

Adopting an onomasiological perspective in describing such general concepts as manner is a first step toward achieving a better understanding of their underlying mechanisms. As argued in Stosic (2011) and Moline and Stosic (2016), the study of manner has to be based on a multilevel approach because there is a considerable diversity of linguistic devices for encoding it within and across languages. The significant body of research that has been carried out to date has mainly focused on two types of linguistic devices involved in the expression of manner, namely syntactic (5) and lexical (6) ones:

- (5) *The boy climbed **nimbly** up the tree.*
- (6) *The bear was **walking/ wandering/ roaming** in the forest.*

Surprisingly, perhaps, three other kinds of devices for encoding manner received relatively little attention in previous research, namely morphological (e.g. *wrong-ly*, *frog-wise*), grammatical (e.g. *how*) and prosodic means (e.g. by using intonation) (Stosic 2011: 131–132). In the subsections that follow, I briefly present these five types of linguistic elements invoked to express manner.

2.1. Syntactic expression of manner

The most common way of expressing manner across languages is to use many kinds of syntactic units and constructions. Generally called “manner adjuncts” or “manner adverbials”, such syntactic constituents are combined with verbs, adjectives or other adverbs in order to indicate a specific way in which a given process, state or quality is realised (see 3–6). In French for example³, according to the extensive research on manner adjuncts (see Guimier 1996: 25; Rémi-Giraud 1998; Patri 1998; Moline 2011; Moline and Stosic 2016), the most frequent types of these syntactic devices are:

- Adverbs

- (7) *La Dionée attrape-mouche digère lentement*
the Venus flytrap digest.PRS.3SG slowly
sa victime.
its victim
‘The Venus flytrap slowly digests its victim’
- (8) *L’eau est délicieusement bonne.*
the water be.PRS.3SG deliciously good

³ On the same topic in English and some other languages, see Hasselgård (2010) and Lang *et al.* (eds) (2003), among others. Moreover, across languages, there is a great variety of linguistic devices for encoding manner, such as ideophones in Japanese (cf. Wienold 1995), serial verb constructions in Thai (cf. Kingkarn 2008), and so on.

‘The water is deliciously good’

- Prepositional phrases

(9) *Caligula se lève avec un effort*

Caligula get.up.PRS.3SG with an effort

visible. (Camus, *Caligula*, p. 27)

visible

‘Caligula stands up with a visible effort’

(10) *Mahiette hochait la tête*

Mahiette shake.IMP.3SG the head

d' un air pensif. (Hugo, *Notre-Dame de Paris*, 1831)

of an air pensive

‘Mahiette shook her head with a pensive air’

- Finite subordinate clauses

(11) *Il est cependant impossible que cela*

it be.PRS.3SG however impossible that it

finisse comme ils l' ont décidé.

end.SUBJ.PRS.3SG as they it have.PRS.3SG decide.PTCP

‘It is, however, impossible that this should end as they have decided’

(12) *Il regardait autour de lui comme s' il*

he watch.IMP.3SG around of him as if he

faisait quelque chose d' interdit.

do.IMP.3SG something of forbidden

‘He was looking around as if he were doing something forbidden’

- Infinitival clauses

(13) *Athos, sans rien dire, vida ses poches.*

Athos without nothing say.INF empty.PST.3SG his pockets

‘Athos, without saying anything, emptied his pockets’

(14) *Mon cœur battait à se rompre.*

my heart beat.IMP.3SG at/to break.INF

‘My heart was beating as if it would break’

- Gerunds

- (15) *Et lord de Winter se retira*
 and lord of Winter himself retire.PST.3SG
en jurant. (Dumas, *Trois Mousquetaires*, 1844)
 in swear. PTCP.PRS
 ‘And Lord de Winter withdrew swearing’
- (16) *Le petit garçon partit en courant.*
 the little boy leave.PST.3SG in run.PTCP.PRS
 ‘The little boy set off at a run’

- Absolute constructions

- (17) *Je peux faire ce gâteau les yeux fermés.*
 I can.PRS.1SG INF this cake the eyes
fermés.
 close.PTCP
 ‘I can make this cake with my eyes shut’
- (18) *Le chef de cabinet entra, des papiers à la main.*
 the chief of staff enter.PST.3SG the papers in the hand
 ‘The chief of staff entered, papers in hand’

- Invariable adjectives

- (19) *Achetez malin!*
 buy.IMP.2PL smart.ADJ
 ‘Buy smart’
- (20) *Comment manger intelligent?*
 how eat.INF intelligent.ADJ
 ‘How can one eat intelligently?’

In all these examples, manner adjuncts appear as – both syntactically and semantically – related to and depending on another element whose value is to be modified. Their role is thus to convey – on the syntagmatic level – a variety of possible ways that a given process, state or quality expressed by the governing element is realized.

2.2. Lexical encoding of manner

It is quite well-known nowadays that languages have a wide range of lexical items whose meaning involves a manner component. For instance, the meaning of the English verb *to run* is defined in the *Cambridge dictionary online* as ‘to move along, faster than walking, by taking quick steps in which each foot is lifted before the next foot touches the ground’. In the same dictionary, the noun *method* is defined as ‘a particular way of doing something’. Consequently, besides syntactic means of expressing manner, what is regularly found in languages is a very large lexicon of verbs (21), simple adverbs (22), and nouns (23) that encode manner in their lexical entry:

- (21) *marcher* ‘to walk’, *errer* ‘to roam’, *nager* ‘to swim’, *boiter* ‘to limp’,
hurler ‘to shout’, *murmurer* ‘to murmur’, *beugler* ‘to bellow’,
grignoter ‘to snack, to nibble’, *dévorer* ‘to devour’, and so on.
- (22) *vite* ‘fast, quickly’, *ainsi* ‘thus, so’, *bien* ‘well’, *mal* ‘bad, badly’
- (23) *manière* ‘manner, way’, *façon* ‘manner, way’, *mode* ‘way, mode’,
méthode ‘method’, *tactique* ‘tactic(s)’, and so on.

Among the above-mentioned word classes, the verb is by far the most widely studied one when talking about the lexical coding of manner. To address the lexicalization of manner in the verbal domain, several approaches have been used. Three of them are of particular interest to the present discussion and will be briefly commented on here.

First, manner is an important component in describing lexical meaning in many approaches postulating some type of predicate decomposition (see Jackendoff 1983, 1990; Rappaport Hovav and Levin 1998; Hale and Keyser 2002; and Levin and Rappaport Hovav 2011 for an overview). As such, manner is assumed to capture some facets of meaning which determine the grammatical behavior of the verb. For instance, in

Rappaport Hovav and Levin's model of lexical decomposition, a verb's meaning is basically described as combining one or more "primitive predicates" (e.g. ACT, BECOME, CAUSE, and so on) with one or more "constants" (e.g. THING, PLACE, MANNER, STATE, and so on), both of which are limited in number. The role of the manner component is to modify primitive predicates, as illustrated in Figure 1 by the example of the manner of motion verb *walk*, and this is what defines the sub-class of manner of motion verbs and distinguishes them from motion verbs such as *go*, *exit* or *leave*, whose meaning does not involve a manner component and that can be considered as neutral ones:

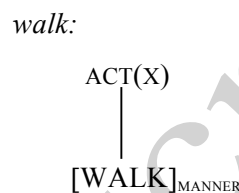


Figure 1. Levin and Rappaport Hovav's (1998: 251) model of lexical decomposition of manner of motion verbs

All manner of motion verbs (e.g. *fly*, *run*, *swim*, *stroll*, *ambulate*) are expected to share this basic lexical semantic template.

Secondly, the lexicalization of manner has been studied in cognitive linguistics and cognitive psychology in extensive cross-linguistic research on the expression of motion, initiated by Talmy's typology (cf. Talmy 2000). In this approach, manner is one of two crucial components for defining possible patterns of lexicalization of motion events across languages. More specifically, Talmy brings to the fore two main lexicalization patterns depending on how two crucial semantic components of motion/location descriptions – path⁴ and manner – are encoded across languages. Thus, by

⁴ This component is defined by Talmy (2000: 25) as "the path followed or site occupied by the Figure object with respect to the Ground object". There are also four

observing the kinds of lexical and syntactic means languages tend to use when expressing path and manner, and their distribution in a single clause, Talmy argues that most languages fall into one of the two main typological groups, namely *Verb-framed languages* (VL) or *Satellite-framed languages* (SL), depending on where the core information (i.e. the path) is expressed. In VL (e.g. French, Turkish, Japanese, Basque, Hebrew), the “path of motion” is preferably encoded by the verb so that the manner generally appears as optional information expressed by some marginal elements (see 24). In SL (e.g. Slavic, English, Dutch, Finnish, Hungarian), the path is preferably encoded by various particles or “satellites” associated with the verb (prepositions, prefixes, postpositions, etc.), while manner is expressed directly in the verb (see 25):

(24) *Rentre vite dans la maison.* (French, VL)

‘Come back quickly into the house.’

(25) *Run back into the house.* (English, SL)

This “division of labor” in expressing path and manner makes the manner component more or less salient across languages. Thus, since in Satellite-framed languages the manner is encoded in the core element of a sentence, it can be considered as linguistically and cognitively much more salient than in Verb-framed languages, where the manner is expressed in a relatively marginal way, or completely omitted (see Slobin 2006; Cardini 2008). Related to this, SL are claimed to have generally a very large lexicon of manner of motion verbs, and speakers of this group of languages to use manner verbs more frequently when describing motion events, as opposed to VL where manner of motion verbs are less common both in the lexicon and

other key elements in motion event descriptions: Figure (the moving or located entity), Ground (the reference object), Motion (“the presence per se of motion or locatedness in the event”), and Cause.

in use (cf. Slobin 1996, 2004; Özçaliskan and Slobin 1999). Two main benefits of this vein of research are that it pointed out that the lexical encoding of manner is widespread across languages, and it made a significant contribution to the indexing of devices and strategies for encoding manner in the motion domain.

The third key approach to the lexicalization of manner was developed by G. Miller and C. Fellbaum in the early nineties in the framework of the *WordNet* project (see Fellbaum and Miller 1990; Miller and Fellbaum 1992; Fellbaum 2002; Fellbaum (ed.) 1998). These scholars showed that manner plays an important role in organizing the verbal lexicon in English. More particularly, they argued that English verbs are mainly structured in terms of “troponymy”, which is a hierarchical “manner” lexical relation: “most lexicalized verb concepts refer to an action or event that constitutes a manner elaboration of another activity or event” (Miller and Fellbaum 1992: 217). The verb *to devour* meaning ‘to swallow or eat up hungrily’ (*WordReference*) can be considered as a troponym of the more general verb *to eat*. Similarly to the relation of hyperonymy, Fellbaum (2002: 24) argues that “the subordinate concept contains the superordinate, but adds some additional semantic specification of its own”. What is interesting in this approach is that all semantic domains are investigated, contrary to the previous approach that only focuses on the motion domain. An in-depth exploration of the *WordNet* database reliably confirms that no conceptual domain resists the lexical encoding of manner.

Taken together, the results of this research in lexical semantics strongly suggest that manner is deeply entrenched in the lexicons of natural languages, and highly present in the verbal domain. It is also obvious that manner affects all semantic domains, without exception.

2.3. Morphological expression of manner

Even though morphological means of expressing manner are rarely investigated by scholars, this way of encoding manner is observed in many languages including Lakhota⁵ (26) (cf. Foley and Van Valin 1984, 39–47), Zulu⁶ (27), Russian (28) (cf. Comrie 1985, 344), French (29) (cf. Amiot and Stosic 2011; Stosic and Amiot 2011, 2019) and Serbian (30) (cf. Stosic 2013b):

- (26) *blečha* ‘break’ > *ya-blečha* ‘break or cut with the teeth’ (Lakhota)
blečha ‘break’ > *na-blečha* ‘break by kicking or stepping on’
- (27) *buza* ‘to ask’ > *buz-isisa* ‘to ask insistently’ (Zulu)
ndiza ‘to fly’ > *ndizandiza* ‘to fly a little’
- (28) *nažat* ‘to press’ > *pri-nažat* ‘to press lightly’ (Russian)
- (29) *sauter* ‘to jump’ > *sautiller* ‘to hop (around)’ (French)
boiter ‘to limp’ > *boitiller* ‘to limp slightly’
voler ‘to fly’ > *voleter* ‘to flutter’
- (30) *skakati* ‘to jump’ > *ska-kut-ati* ‘to hop (around)’ (Serbian)
trčati ‘to run’ > *trč-kara-ti* ‘to run around’
hramati ‘to limp’ > *hram-uck-ati* ‘to limp slightly’

In all these examples, due to affixation, the derived items express a specific way of performing the action indicated by the verb stem.

Morphological encoding of manner is essentially found in two word classes: adverbs and verbs. In many languages, manner adverbs are formed by different morphological operations taking adjectives as base, as is the case in English (31) and French (32):

- (31) *beautiful-ly*, *calm-ly*, *dangerous-ly*, *crab-wise*, *prayer-wise*...
- (32) *étrange-ment* ‘strangely’, *rapide-ment* ‘quickly’, *fidèle-ment* ‘faithfully’...

⁵ The Lakhota language is spoken in North Central USA and in Canada.

⁶ Zulu is spoken in South Africa by about 10 million native speakers.

As for verbs, very often (see 27–30), but not exclusively (see 26), the manner component results from the use of a different kind of morphological processes which belong to what is referred to as “evaluative morphology” and/or “pluractionality” (see Stosic and Amiot 2019 for an overview). As shown in examples (27–30), these markers generally involve a substantial modification in the realization of events described by the verb stems, especially in their internal structure (see among others Cusic 1981; Newman 1990, 2012; van Geenhoven 2004; Grandi 2009; Greenberg 2010; Tovenia 2010a, b; Amiot and Stosic 2011). For instance, the French evaluative verb *voleter* ‘to flutter’ (< *voler* ‘to fly’) is used to describe actions of flying that are performed in a non-canonical way, that is actions when a flying animal makes a series of quick random movements up and down or from side to side. This semantic specification directly results from evaluative marking. Due to this modification, the action is represented as occurring in a specific way, and hence as not conforming to its prototypical realization involved by the base. Stosic and Amiot (2011, 2019) argue that this deviation from the norm is what generates a manner interpretation of the morphologically complex verbs at issue.

2.4. Grammatical encoding of manner

Manner is also encoded across languages on a highly abstract level of representation, which strongly supports a possible grammatical nature of this concept. What warrants this claim is the existence, in the vast majority of world languages, of a small set of linguistic elements belonging to closed inventories and conveying manner. Thus, one can generally find in languages a series of interrogative and indefinite adverbs or pronouns like, in English, *how* (33), *anyway*, *anyhow*, *someway*, *somehow* (34), *no way*, or in French,

comme ‘like, as’ (35) and *comment* ‘how’ whose role is to mark the manner variable in an abstract way:

- (33) *A boy trained among men would never have dreamed of skinning a ten-foot tiger alone, but Mowgli knew better than anyone else **how** an animal’s skin is fitted on, and **how** it can be taken off.* (R. Kipling, *The Jungle Book*, 1894)
- (34) *She liked the word “hysterics” and felt **somehow** as if it had an effect on him.* (F.H. Burnett, *The Secret Garden*, 1911)
- (35) *Il est mort **comme** on ne meurt plus.* (J. Brel, *La Statue*)
 he be.PRS.3SG die.PTCP like one/we not die.PRS.3SG more/no.longer
 ‘He died like we no longer die’

2.5. Suprasegmental encoding of manner

There are languages that use tonal and prosodic devices for encoding manner. This generally happens in tone languages where the type and the mobility of the tone may play a discriminating role at the semantic level. In such languages, a manner adverbial can be marked by a tone shift or by a change in the type of tone/accent. For instance, in Serbian, which has four “accents” involving either a rising or falling tone on either long or short vowels, two different accents enable speakers to make a distinction in some cases between adjectives and the corresponding manner adverbs. Thus, the expression *mâlo* ‘little’ is characterized by a long vowel /a/ with falling tone in (36) and is to be analyzed as a noun, that is as a substantivated neuter adjective, whereas in (37) where the same syllable integrates a short vowel with falling tone, *mâlo* ‘a little bit’ corresponds to manner adverb:

- (36) *Mâlo se pomerilo.*
 SUBS-little REFL V-move-PST.PTCP.N.SG

- ‘Something little moved’
- (37) *Màlo se pomerilo.*
 ADV-little REFL V-move- PST.PTCP.N.SG
 ‘(Something) moved a little bit’

Another type of suprasegmental device for encoding manner is found in Diyari, a language spoken in Australia. According to Patri (1998: 153), in this language, the insertion of a prosodic break after an adjective integrated in the subject NP (38) turns it into the corresponding manner adverb (39):

- (38) *nawu-ja ηanti tuŋka pani-ji*
 PR-3.SG SUBS-meat ADJ-rotten V-smell-3.SG.PRS
 ‘this rotten meat smells’
- (39) *nawu-ja ηanti | tuŋka pani-ji*
 PR-3.SG SUBS-meat ADV-rotten V-smell-3.SG.PRS
 ‘this meat smells rotten’

Adopting an onomasiological approach to manner has allowed me to point out five different types of linguistic devices used for encoding this undeniably complex notion. This means that the study of manner must rely on a multilevel approach that takes this diversity of linguistic strategies into account. However, one might legitimately ask whether this opens the way to a better understanding of manner, and if so, in which way. An in-depth analysis of how these devices operate shows that, in spite of their diversity, all of them work in a similar way. Consequently, elucidating the main manner-generating mechanisms that operate at different linguistic levels is a key way to better understand this concept.

3. Looking for a more operational definition of manner

The second imperative for describing general concepts like manner is to point to mechanisms and/or features common to all means of its expression. I thus assume that there exists a limited set of mechanisms for processing manner interpretation, operating either in language or in language use, whatever the devices at work. The aim of this section is to isolate and bring together various mechanisms involved in the processing of manner in order to achieve both a more precise and a holistic definition of this semantic concept.

3.1. Subsidiarity of manner, or the need for a substrate conceptual content

First of all, processing manner both in language and language use is closely dependent on the existence of a substrate conceptual content that is to be modulated (but not altered in its essence) by various characterizing features and/or parameters. This mechanism is fundamental for describing the concept of manner, which unambiguously acts as an operator of modification. This can be schematized as follows:

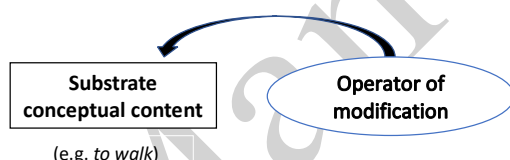


Figure 2. Manner as a modifying mechanism

This is what could explain why manner is commonly considered as a non-autonomous semantic value: the conceptual subsidiarity of manner requires it to be never realized separately and apart from some other semantic content (see also Talmy 2000: vol. 2: 37). This fits perfectly with the claim that manner is an accompanying or additional aspect of an action, or even a circumstance concept par excellence as claimed by traditional grammar.

The modification at issue can be processed on different linguistic levels, which means that the substrate content can be modified syntactically, lexically, morphologically, and so on. For instance, the concept of walk can be processed:

- on the syntactic level by so-called manner adjuncts as in (40):

(40) *She was **quietly** walking towards the door.*

- on the lexical level by adding semantic specifications to a more general predicate as in (41), where a general motion predicate (*move*) acts as the substrate element that is lexically processed by a manner modifier (*unsteadily or with difficulty*):

(41) *to limp* (vs. *to walk*) ‘moving unsteadily or with difficulty’ (CED)

- on the morphological level by specific patterns of word formation as in (42):

(42) *march-**ott**-er* (< *marcher* ‘to walk’) ‘to walk with difficulty, to walk taking small steps, to walk unsteadily’

The same could be said about grammatical and suprasegmental levels. In the light of these illustrations, Figure 2 can be completed as follows:

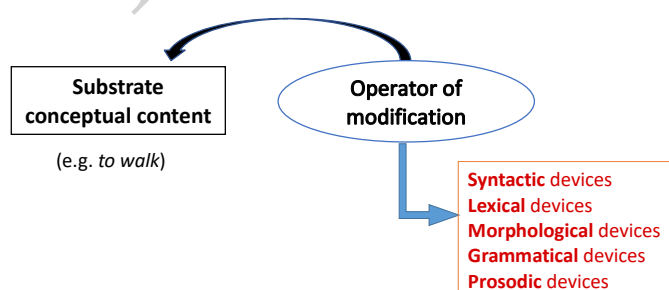


Figure 3. Different kinds of modifying devices

3.2. Constraints on the substrate: actions, states, qualities

The second key feature for understanding manner is that the nature of the substrate is ontologically constrained: a manner interpretation can only appear when the substrate element belongs to the categories of actions (43), states (44) or qualities (45):

- (43) *pleurer timidement* ‘to cry timidly’
sangloter ‘to sob, to weep with a catching of the breath or in sudden, short gasps’ (WR)
- (44) *rester assis la tête appuyée sur sa main* ‘to rest seated with his/her head on his/her hand’
- (45) *une voix joliment gutturale* ‘a beautifully guttural voice’

This characteristic of manner has been particularly highlighted in very extensive research on manner adverbs, which are claimed to modify essentially verbs (=actions, states), adjectives (=qualities) or adverbs (=qualities). This constraint is crucial for discriminating between manner and quality, the latter being efficient when what is modified does not belong to actions, states or qualities, but to some other ontological category (e.g. substances and/or individuals). Thus, the same attribute can be interpreted as meaning manner (46) or quality (47) depending on the nature of (what is referred to by) its governor:

- (46) *Plusieurs chevaux couraient en liberté.*
many horses run.IMP.3.PL in liberty
‘Many horses were running free’
- (47) *nous avons vu beaucoup d’ animaux en liberté.*
we have.PRS.1.PL see.PTCP a.lot. of animals
in liberty
‘We saw a lot of wild animals’

These examples show that when applied to a verb (*courir* ‘run’), *en liberté* ‘free’ expresses a specific way of performing the action of running whereas it marks a quality of individuals (cf. Strawson 1959) when its governor is a nominal expression (*animaux* ‘animals’). This semantic and conceptual switch can be explained by the fact that both manner and quality fall within the same cognitive operation of characterization, which consists in assigning an attribute to an entity. The proximity of quality and manner is well-known in linguistics and rather well documented by several scholars (see among others Le Goffic 2002; Van de Velde 2009; Stosic 2013a).

3.3. *Manner as diversification*

The third key element in defining the concept of manner is to be found in the way the modification is made. More precisely, the modifying role of manner is tightly related to the operation of diversification, which is crucial for understanding the way in which manner is processed in language and language use. As I argued in Stosic (2019: 153):

“The role of manner thus consists in diversifying a given substrate by specific qualitative features, i.e. in distinguishing actions, states or qualities of the same nature from each other (see Van de Velde 2009; Stosic 2011). This process leads to splitting a given class of actions, states or qualities into sub-sets of referents that, while belonging to the general class at issue, are somehow different due to the presence of some distinctive feature.”

This is what can be observed in examples (48–50) illustrating three possible ways of diversifying the action of eating:

- (48) *Les enfants mangent le gâteau avec beaucoup*
the children eat.PRS.3.PL the cake with a lot
de plaisir.

of pleasure

‘The children are eating the cake with great pleasure’

(49) *Les enfants dévorent le gâteau.*

the children devour.PRS.3.PL the cake

‘The children devour the cake’

(50) *Regarde comme ils mangent le gâteau.*

look. IMP.2.SG like they eat.PRS.3.PL the cake

‘Look how they are eating the cake’

More specifically, each example suggests a particular instantiation of the action of eating. Saying (48) makes this occurrence of the action of eating different both from its prototypical realization (*the children are eating the cake*) and from a range of its other possible realizations (e.g. *the children are eating the cake slowly/ properly/ like little piglets...*). The same could be said for the verb *dévorer* ‘devour’ in (49) that entails diversification of the same action on the lexical level as suggested by its lexicographic definition: ‘to eat something eagerly and in large amounts so that nothing is left’ (*CED*). The general predicate (*to eat*) acts as the substrate element that is lexically processed, whereas the role of the manner component (*eagerly and in large amounts so that nothing is left*) consists in diversifying, and thereby in modifying, the root predicate. As long as *eat* is used without any manner determination, it describes a prototypical representation of the given action. By using a verb such as *devour* or manner adjuncts such as *slowly*, *with great pleasure* and so on, speakers stress some specific, probably dominant, distinctive aspects of the occurrence of the action of eating that they talk about. Consequently, each manner determination, be it syntactic, lexical, morphological or grammatical, reflects individual variations of a given substrate content. The same applies to the modification of qualities as seen in the following examples, where the properties of being different (51) and mild (52) are diversified respectively by the adverb *joliment* ‘nicely’ and the adverb *délicatement* ‘delicately’:

- (51) *Ce détail vous rend **joliment** différente.*
 this detail you make.PRS.3.SG nicely different
 ‘This detail makes you nicely different’
- (52) *Le goût de ce fromage est **frais** et **délicatement** doux.*
 the taste of this cheese be. PRS.3.SG fresh and
 delicately mild
 ‘The taste of this cheese is fresh, and delicately mild’

3.4. Compositionality of the concept of manner

The fourth peculiarity of the concept of manner is that it is by no means monolithic; it is rather a complex concept made up of a wide variety of more basic semantic values and/or parameters whatever the level that it operates on. Thus, comparing possible interpretations of manner adjuncts like the ones in examples (53) and (54) clearly shows that a wide range of semantic values, all subsumable under the concept of manner, can be distinguished on the syntactic level. More precisely, while all the adjuncts in bold can answer the question with *how*, it is obvious that in (53) “with fingers” and “with all the more pleasure” do not involve the same meaning (instrument vs attitude). Moreover, in (54) “delicately” and “with parsley and garlic” are both suitable as manner adjuncts even though they diversify the action of seasoning in two different ways (quality of action vs special kind of instrument).

- (53) *Les enfants mangeaient le gâteau **avec les doigts**,*
 the children eat.IMP.3.PL the cake with the fingers
*mais **avec d’autant plus de plaisir.***
 but with all the more of pleasure
 ‘The children were eating the cake with their fingers, but with all the more pleasure’

- (54) *Le cuisinier a délicatement assaisonné*
 the cook have.PRS.3.SG delicately season.PTCP
le plat avec du persil et de l'ail.
 the dish with the parsley and the garlic
 'The cook delicately seasoned the dish with parsley and garlic'

As for the lexical level, one can easily see that, for example, not all manner of motion verbs involve the same semantic specification, as is clearly suggested by the contrast between the meanings of *to limp*, *to run*, *to zigzag* and *to bike* in (55a–d):

- (55) a. *to limp* 'to move unsteadily or with difficulty'
 b. *to run* 'to move along, faster than walking, by taking quick steps in which each foot is lifted before the next foot touches the ground'
 c. *to zigzag* 'to move by going first in one direction and then in a different direction, and continuing in this way'
 d. *to bike* 'to go somewhere by bicycle'

As I argued in my previous work dealing with manner of motion verbs (see Stosic 2009, 2019) and with manner of speech verbs (see Moline and Stosic 2016), on the lexical level too, manner should be seen as a cluster concept, rather than a unitary semantic category. Indeed, an in-depth semantic analysis of more than 500 manner of motion verbs in French has shown that their manner component is built on the basis of a restricted set of more elementary features such as: BODY MOTION PATTERN (*to walk*, *to stagger*), SPEED (*to run*, *to rush*, *to dawdle*), SHAPE OF THE PATH (*to zigzag*, *to weave*), PURPOSE(LESS) (*to wander*, *to roam*, *to weave*), INSTRUMENT (*to paddle*, *to ski*), and so on (see also Cardini 2008; Malt et al. 2008, 2010; Slobin et al. 2014). In some cases, the manner component is triggered by one of these parameters, in others it is underlain by the combination of two of them. These basic semantic features can be shared by several lexical items from the same semantic domain, as

well as from different domains. For instance, the parameter of PURPOSE(LESS) is also relevant in the domain of speech verbs (e.g. *to prattle, to ramble*) (cf. Moline and Stosic 2016). This suggests that these features are probably numerous but certainly limited in number.

Finally, the possibility of decomposing manner into more basic values also appears on the morphological level. According to Amiot and Stosic (2011) and Stosic and Amiot (2011), evaluative and pluractional readings specific to some verbs are supported by a range of semantic values such as diminutive, augmentative, iterative, conative, incassative ones, and so on.

Manner thus appears as semantically compositional by nature whatever the level it occurs on. This semantic complexity of the concept of manner has been labelled by many scholars as its heterogeneity and multifacetedness (cf. Ikegami 1969; Snell-Hornby 1983; Guimier 1996; Slobin 2004: 255; Caballero 2007; Stosic 2009). A possible articulation of manner with all these semantic values will be addressed in the fourth section below.

The previously mentioned difficulties regarding the lack and fuzziness of definition of manner can be overcome by grouping together the mechanisms and features pointed out in this section. Such an attempt was already made in Stosic (2011, 2013a), and in Moline and Stosic (2016), and the following definition was proposed:

“Manner is a complex semantic value, incidental by nature to some substrate element that is processed by various lexical, syntactic, morphological, grammatical and prosodic means and strategies. This processing results in diversifying the substrate by specific qualitative features, and thereby in characterizing/modulating it. The substrate must belong to one of the following ontological types: actions, states or qualities.” (Stosic 2019: 152)

The main advantage of this definition lies in the fact that it provides a unified framework for the study of the concept of manner whatever linguistic means it is expressed by. It also offers a first set of the main distinctive features and mechanisms peculiar to this semantic concept, as well as their articulation, with the aim of better understanding manner, and of opening new perspectives to more accurately address linguistic data related to it, in all their diversity and in all their complexity.

4. Manner and some neighbouring concepts

As said in the introductory section of the paper, an accurate definition of manner should provide appropriate criteria for either its delineation from many contiguous semantic values or its possible articulation with them. Although the definition proposed in the previous section enables manner to be distinguished from quality, for example, it does not make a clear distinction from or articulation with a range of notions (e.g. instrument, means, comparison, etc.) whose relationships with manner remain somewhat problematic (see §§ 1.3; 3.4).

In order to clarify the relationships between manner and these related notions, Moline and Stosic (2016: 189-192) proposed to distinguish between *manner in a broad sense* and *manner in a narrow sense*, arguing that manner is to be considered as a two-level concept, which operates on two different levels of abstraction. At the highest level of abstraction, manner corresponds to an extremely general concept that comprises a great diversity of semantic values from a lower level of abstraction. Indeed, at the lower level, to begin with syntax, one can find in addition to adjuncts expressing manner in a narrow sense (56), those expressing instrument (57), means (58), comparison (59), and even, in an ad hoc manner, intensity (60) or temporality (61):

- (56) *Jany lit attentivement son livre.*

- Jany read.PRS.3SG carefully her book
 ‘Jany is perusing her book’
- (57) *Il a brutalement battu son chien*
 he have.PRS.3SG brutally beat.PTCP his dog
avec un bâton de base-ball.
 with a bat of base-ball
 ‘He has brutally beaten his dog with a baseball bat’
- (58) *Le paysage audiovisuel s’ est enrichi par le lancement des radios privées.*
 the landscape audio-visual itself be.PRS.3SG
 enrich.PTCP by the launching of.the radios private
 ‘The audio-visual sector has been enhanced through the launching of private radio stations’
- (59) *J’ ai décidé d’ être courageux*
 I have.PRS.1SG decide.PTCP to be.INF brave
comme mon grand frère.
 like my big brother
 ‘I decided to be brave like my big brother’
- (60) *Mon cœur battait très fort.*
 my heart beat.IMRFT.3SG very hard
 ‘My heart was beating very strongly’
- (61) *Sans amour, la vie passe en un éclair.*
 without love the life pass.PRS.3SG in one flash
 ‘Unless you love... your life will flash by’ (movie title)

In all these cases, shifting from the basic values conveyed at the lower level by these constituents to the general concept of manner could be explained by the fact that they all act as operators of diversification by carrying out a qualitative modification of a process or of a state (compare *to read* vs *to read carefully*, *to beat* vs *to beat strongly*). This is precisely what triggers a twofold interpretation of such adjuncts. One of the major benefits of the two-level analysis is that it reconciles the simultaneous presence of two semantic

interpretations (e.g. manner and instrument, or manner and means, etc.), and keeps them as equally relevant by distributing them onto two different levels of abstraction. The same applies to the lexical and morphological levels, where a limited set of basic semantic values from the lower level (see § 3.4 above) generates the manner interpretation at the higher level. The hierarchical representation of manner in Figure 4 seeks to illustrate the two-level structure of this general concept:

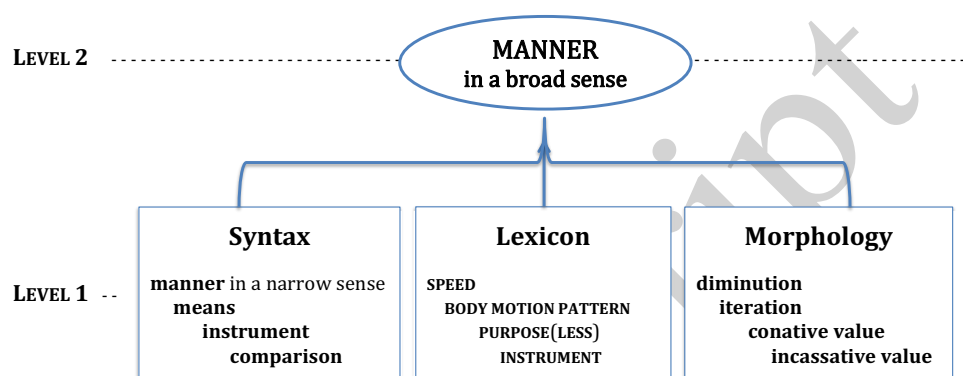


Figure 4. Manner as a two-level concept

This two-level conception of the concept of manner also explains why interrogatives such as *how* in English, or *comment* ‘how’ in French, that belong to the highest level of abstraction, can be used to question both about manner and many other more specific semantic values (instrument, means, comparison, etc.).

Conclusion

The paper has a twofold aim: to draw up a summary of previous attempts at defining manner, and to propose a more operational definition of this concept by considering, in a holistic approach, the entire range of linguistic means used to express it. First, I discussed the fact that manner remains an ill-defined notion in linguistics, despite a wealth of research especially in syntax, lexical semantics, and typological studies. Next, I

recalled recent advances in the study of manner showing that languages employ not only syntactic devices for encoding manner, but also lexical, morphological, grammatical and suprasegmental ones. Underlying this diversity, one can find a set of features and mechanisms at work whenever the manner component appears in the interpretation. These elements common to all manner expressions make it possible to propose a more comprehensive definition of manner that aims to cover the study of all the means of expression whatever the level they originate from (lexicon, syntax, morphology, etc.). In this light, manner is defined as an operator of diversification of actions, states and qualities, whose role is to stress the way a given occurrence of these three kinds of entities is qualitatively distinguishable from both their prototypical and their other possible instantiations.

An important aspect of defining manner is that it should not be viewed as a unitary, indecomposable notion but rather as a cluster concept built up from a limited set of semantic values that contribute to the construction of the manner interpretation, both in language and language use. I thus claim, based on Moline and Stosic (2016), that manner is a two-level concept operating at different levels of abstraction. This conception provides a valuable explanation of the clustering nature of manner, and of the fact that it encapsulates a wide range of semantic notions that, at first sight, make manner extremely heterogeneous. This way of approaching manner makes it possible to both delineate it from such notions as quality or quantity, and to articulate it with a series of other neighbouring semantic values, whose relationships to manner have remained ill-defined in previous research on the topic.

Abbreviations

1	first person
2	second person
3	third person

ADJ	adjective
ADV	adverb(ial)
IMP	imperative
INF	infinitive
PR	pronoun
PL	plural
PRS	present
PST	past
PTCP	participle
REFL	reflexive
SG	singular
SUBS	substantive

References

- Amiot, D., & Stosic, D. 2011. *Sautiller, voleter, dansoter*: évaluation, pluriactionnalité, aspect. In E. Arjoca-Ieremia, C. Avezard-Roger, J. Goes, E. Moline, & A. Tihu (Eds.), *Temps, aspect et classes de mots: études théoriques et didactiques* (277–297). Arras: Artois Presses Université.
- Bonami, O., Godard, D., & Kampers-Manhe, B. 2004. Adverb Classification. In F. Corblin, & H. d. Swart (Eds.), *Handbook of French Semantics* (142–185). Stanford: CSLI.
- Caballero, R. 2007. Manner-of-motion verbs in wine description. *Journal of Pragmatics*, 39, 2095–2114.
- Cardini, F.-E. 2008. Manner of motion saliency: An inquiry into Italian. *Cognitive Linguistics*, 19(4), 533–569.
- Comrie, B. 1985. Causative verb formation and other verb-deriving morphology. In T. Shopen (Ed.), *Language typology and syntactic description* (309–348). Cambridge: Cambridge University Press.
- Cusic, D. 1981. *Verbal Plurality and Aspect*. PhD dissertation: Stanford University.
- Fellbaum, C. (Ed.) 1998. *WordNet. An Electronic Lexical Database*. Cambridge, MA : MIT Press.
- Fellbaum, C. 2002. On the Semantics of Troponymy. In R. Green, C. Bean & S. Hyon Myaeng (Eds.), *The Semantics of Relationships: An Interdisciplinary Perspective* (23–34). Dordrecht: Kluwer.

- Fellbaum, C., & Miller G. 1990. Folk psychology or semantic entailment ? A reply to Rips and Conrad. *Psychological Review*, 97, 565–570.
- Foley, W. A., & Van Valin, R. D. 1984. *Functional syntax and universal grammar*. Cambridge: Cambridge University Press.
- Gary-Prieur, M.-N. 1982. ‘Adverbe de manière’: que signifie cette étiquette ?. *Lexique*, 1, 13–23.
- van Geenhoven, V. 2004. *For-Adverbials, Frequentative Aspect, and Pluractionality*. *Natural Language Semantics*, 12, 135–190.
- Grandi, N. 2009. Restrictions on Italian Verbal Evaluative Suffixes: The Role of Aspect and Actionality. *York Papers in Linguistics Series 2*, 10, 46–66.
- Greenberg, Y. 2010. Event Internal Pluractionality in Modern Hebrew: A Semantic Analysis of One Verbal Reduplication Pattern. *Brill’s Annual of Afroasiatic Languages and Linguistics*, 2(1), 119–164.
- Gross, M. 1990. *Grammaire transformationnelle du français. Syntaxe de l’adverbe*. Paris: ASSTRIL.
- Guimier, C. 1996. *Les adverbes du français: le cas des adverbes en –ment*. Paris & Gap: Ophrys.
- Hale, K., & Keyser, S. J. 2002. *Prolegomenon to a Theory of Argument Structure*. Cambridge, MA: MIT Press.
- Hasselgård, H. 2010. *Adjunct adverbials in English*. Cambridge, MA: Cambridge University Press.
- Heine, B., Claudi, U., & Hünnemeyer, F. 1991. *Grammaticalization: A Conceptual Framework*. Chicago: University of Chicago Press.
- Haspelmath, M. 1997. *Indefinite Pronouns*. Oxford: Clarendon Press.
- Iacobini, C. & Vergaro, C. 2012. Manner of motion verbs in Italian: semantic distinctions and interlingual comparisons. In *Atti del XLII Congresso della Società di Linguistica Italiana* (71–87), Roma: Bulzoni.
- Ikegami, Y. 1969. *The semological structure of the English verbs of motion*. Yale University: New Haven, CT.
- Jackendoff, R. 1983. *Semantics and Cognition*. Cambridge: MIT Press.
- Jackendoff, R. 1990. *Semantic Structures*. Cambridge: MIT Press.

- Kingkarn, T. (2008). Verb serialization as a means to express complex events in Thai. In B. Lewandowska-Tomaszczyk (Ed.), *Asymmetric Events* (103-120). Amsterdam/Philadelphia: John Benjamins.
- Lang, E., Maienborn, C., Fabricius-Hansen, C. (Eds.) 2003. *Modifying adjuncts*. Berlin/New York: Mouton de Gruyter.
- Le Goffic, P. 2002. Marqueurs d'interrogation/ indéfinition/ subordination: essai de vue d'ensemble. *Verbum*, 24(4), 315–340.
- Le Goffic, P. 2007. Les mots *qu-* entre interrogation, indéfinition et subordination. *Lexique*, 18, 13–46.
- Levin, B., & Rappaport Hovav, M. 1992. The lexical semantics of verbs of motion: The perspective from unaccusativity. In I. M. Roca (Ed.), *Thematic structure: Its role in grammar* (247–269). Berlin: Foris.
- Levin, B., & Rappaport Hovav, M. 1998. Morphology and Lexical Semantics. In A. Spencer, & A. Zwicky (Eds.), *Handbook of Morphology* (248–271). Oxford: Oxford, Blackwell.
- Levin, B., & Rappaport Hovav, M. 2011. Lexical Conceptual Structure. In P. Portner, C. Maienborn & K. von Stechow (Eds.), *An International Handbook of Natural Language Meaning* (420–440). Berlin: Mouton de Gruyter.
- Malt, B. C., Gennari, S., Imai, M., Ameel, E., Tsuda, N. & Majid, A. 2008. Talking about walking: Biomechanics and the language of locomotion. *Psychological Science*, 19, 232–240.
- Malt, B., Gennari, S., & Imai, M. 2010. Lexicalization patterns and the world-to-words mapping. In B. Malt & P. Wolff (Eds.), *Words and the mind: How words capture human experience* (25–57). New York: Oxford University Press.
- Melis, L. 1983. *Les circonstants et la phrase: étude sur la classification et la systématique des compléments circonstanciels en français moderne*. Louvain: Presses universitaires de Louvain.
- Miller, G., & Fellbaum, C. 1992. Semantic networks of English. In B. Levin & S. Pinker (eds), *Lexical & Conceptual Semantics* (197–229). Oxford: Blackwell.

- Moline, E. 2011. Comment construire un paradigme des ‘compléments de manière’?. In F. Hrubaru & E. Moline (Eds.), *La construction d’un paradigme, Recherches ACLIF* (75–96). Cluj : Echincox.
- Moline, E., & Stosic, D. 2016. *L’expression de la manière en français*. Paris: Ophrys.
- Molinier, C. 1990. Une classification des adverbes en *-ment*. *Langue française*, 88, 28–40.
- Molinier C., & Lévrier, F. 2000, *Grammaire des adverbes. Description des formes en -ment*. Genève & Paris: Droz.
- Newman, P. 1990. *Nominal and verbal plurality in Chadic*. Dordrecht: Foris.
- Newman, P. 2012. Pluractional verbs: An overview. In P. Cabredo Hofherr & B. Laca (Eds.), *Verbal Plurality and Distributivity* (185–209). Berlin: de Gruyter.
- Nilsson-Ehle, H. 1941. *Les adverbes en -ment compléments d’un verbe en français moderne*. Copenhagen: Munksgaard.
- Nøjgaard, M. 1992, 1993, 1995. *Les adverbes français: Essai de description fonctionnelle* (3 volumes). Copenhagen: Munksgaard.
- Nølke, H. 1990. Recherches sur les adverbes : bref aperçu historique des travaux de classification. *Langue française*, 88, 117–127.
- Özçaliskan, Ş., & Slobin, D. I. 1999. Learning ‘how to search for the frog’: Expression of manner of motion in English, Spanish, and Turkish. In A. Greenhill, H. Littlefield & C. Tano (Eds.), *Proceedings of the 23rd Annual Boston University Conference on Language Development* (541–552). Somerville, MA: Cascadilla Press.
- Patri, S. 1998. Fondements indiciels de la syntaxe du complément circonstanciel. In S. Rémi-Giraud & A. Roman (Eds.), *Autour du circonstant* (139–172). Lyon: PUL.
- Rappaport Hovav, M., & Levin, B. 1998. Building Verb Meanings. In M. Butt & W. Geuder (Eds.), *The Projection of Arguments* (97–134). Stanford, CA: CSLI Publications.
- Rémi-Giraud, S. 1998, Le complément circonstanciel. Problèmes de définition. In S. Rémi-Giraud & A. Roman (Eds.), *Autour du circonstant* (65–115). Lyon: PUL.

- Sechehaye, A. 1926. *Essai sur la structure logique de la phrase*. Paris: Champion.
- Slobin, D. I. 1996. Two Ways to Travel: Verbs of Motion in English. In M. Shibatani & S. Thompson (Ed.), *Grammatical Constructions* (195–221). Oxford: Clarendon Press.
- Slobin, D. I. 2004. The many ways to search for a frog: Linguistic typology and the expression of motion events. In S. Strömquist & L. Verhoeven (Eds.), *Relating events in narrative: Typological and contextual perspectives* (219–258). Mahwah, NJ: Lawrence Erlbaum Associates.
- Slobin, D. I. 2006. What makes manner of motion salient? Explorations in linguistic typology, discourse, and cognition. In M. Hickmann & S. Robert (Eds.), *Space in languages: Linguistic systems and cognitive categories* (59–81). Amsterdam & Philadelphia: John Benjamins.
- Slobin, D. I., Ibarretxe-Antuñano, I., Kopecka, A. & Majid, A. (2014). Manners of human gait: a cross-linguistic event-naming study. *Cognitive Linguistics*, 25(4), 701–741.
- Snell-Hornby, M. 1983. *Verb descriptivity in German and English: A contrastive study in semantic fields*. Heidelberg: Carl Winter.
- Stosic, D. 2009. La notion de «manière» dans la sémantique de l'espace. *Langages*, 175, 103–121.
- Stosic, D. 2011. Le sens de manière comme critère de définition d'un paradigme. In F. Hrubaru & E. Moline (Eds.), *La Construction d'un paradigme, Recherches ACLIF* (117–142). Cluj: Echinoc.
- Stosic, D. 2013a. A la recherche du complément de manière prototypique. *SCOLIA* 27, 53–74.
- Stosic, D. 2013b. Manner of motion, evaluative and pluractional morphology. *Oslo Studies in Language*, 5, 61–89.
- Stosic, D. 2019. Manner as a cluster concept: What does lexical coding of manner of motion tell us about manner?. In M. Aurnague, M. & D. Stosic (Eds.), *The Semantics of Dynamic Space in French. Descriptive, experimental and formal studies on motion expression* (141–177). Amsterdam/Philadelphia: John Benjamins.
- Stosic, D., & Amiot, D. 2011. Quand la morphologie fait des manières: les verbes évaluatifs et l'expression de la manière en français. In D. Amiot, W. De

- Mulder, E. Moline & D. Stosic (Eds.), *Ars Grammatica. Hommages à Nelly Flaux* (403–430). Bern: Peter Lang.
- Stosic, D., & Amiot, D. 2019. Motion verbs and evaluative morphology. In M. Aurnague, & D. Stosic (Eds.), *The Semantics of Dynamic Space in French. Descriptive, experimental and formal studies on motion expression* (179–215). Amsterdam/Philadelphia: John Benjamins.
- Strawson, P. F. 1959. *Individuals: An Essay in Descriptive Metaphysics*. London: Methuen.
- Talmy, L. 1985. Lexicalization patterns: Semantic structure in lexical forms. In T. Shopen (Ed.), *Language Typology and Syntactic Description. Vol. 3: Grammatical Categories and the Lexicon* (57–149). New York: Cambridge University Press.
- Talmy, L. 2000. *Toward a Cognitive Semantics*. Cambridge MA: MIT Press.
- Tovena, L. 2010a. Pluractionality and the unity of the event. In M. Aloni & K. Schulz (Eds.), *Amsterdam Colloquium 2009 LNAI 6042* (465–473). Heidelberg: Springer.
- Tovena, L. 2010b. When small is many in the event domain. *Lexis*, 6, 41–58.
- Van De Velde, D. 2009. *Comment, manières d’être et manières de faire. Travaux de Linguistique*, 58, 39–61.
- Virtanen, T. (2008). Adverbials of ‘Manner’ and ‘Manner Plus’ in Written English: Why Initial Placement?. *SKY Journal of Linguistics*, 21, 271–293.
- Westerhoff, J. (2005). *Ontological Categories: Their Nature and Significance*. Oxford, Oxford University Press.

Dictionaries

CED (Cambridge English Dictionary), online edition.

(<https://dictionary.cambridge.org>)

TLFi (Trésor de la langue française informatisé), online edition, Nancy: ATILF.

(<http://atilf.atilf.fr/tlfi>)

WR – WordReference Online Language Dictionaries

(<https://www.wordreference.com>).